
Student Academic Performance Prediction

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1. Problem Description

The Student Performance Prediction System is a machine learning-based project used to predict students' academic performance using factors like attendance, study hours, previous marks, assignments, and participation.

The system analyzes student data to predict outcomes such as pass/fail status or final grades. Its main goal is to help teachers identify weak students early and provide proper academic support.

Technologies used include Python, Pandas, NumPy, and machine learning algorithms like Logistic Regression, Decision Tree, and Random Forest.

1.1 Dataset Description

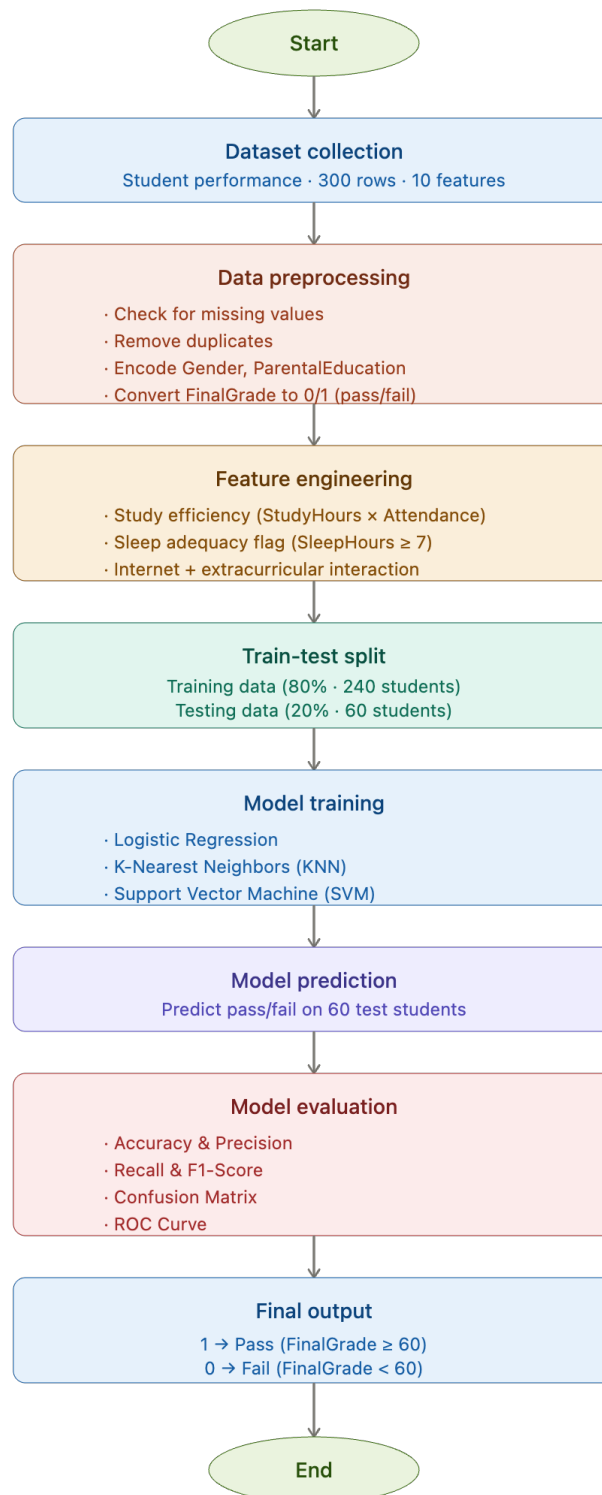
The dataset used in the Student Performance Prediction System contains information about students and their academic-related factors. It consists of **300 records** and **10 attributes**.

Attributes in Dataset

Attribute	Description
Age	Age of the student
Gender	Gender of the student
StudyHours	Number of study hours per day
Attendance	Attendance percentage
ParentalEducation	Education level of parents
InternetAccess	Availability of internet access (0/1)
Extracurricular	Participation in extracurricular activities (0/1)
SleepHours	Average sleep hours per day
PreviousGrade	Previous academic grade/marks
FinalGrade	Final grade obtained by the student

The dataset is used to analyze the factors affecting student performance and to predict the final academic result using machine learning algorithms.

1.2 Flow Diagram



2. Modules for Project Implementation

2.1 Modules Used(mention all the inbuilt/ custom modules used and the purpose)

Random Forest Classifier	ML Module	Used to improve prediction accuracy
Train_Test_Split	Custom Function from Scikit-learn	Used to divide dataset into training and testing data
Accuracy Score	Evaluation Module	Used to calculate model accuracy
Seaborn	Inbuilt Python Library	Used for statistical data visualization
Logistic Regression	ML Module	Used for prediction of student performance
NumPy	Inbuilt Python Library	Used for numerical operations and array handling
Scikit-learn	Inbuilt Python Library	Used for machine learning algorithms and model evaluation
StandardScaler	Custom Module from Scikit-learn	Used for feature scaling and normalization
Matplotlib	Inbuilt Python Library	Used for data visualization and plotting graphs
Pandas	Inbuilt Python Library	Used for data loading, cleaning, and manipulation
Decision Tree Classifier	ML Module	Used for classification and prediction
Module	Type	Purpose

2.2 Platform Used:

Platform	Reason for Using
Python	Main programming language used for implementing machine learning algorithms
Jupyter Notebook	Used for writing, testing, and executing Python code interactively
Anaconda	Provides Python environment and required data science libraries
VS Code	Used for code editing and project development
Windows Operating System	Platform used for running the project and required tools

3. Machine Learning Model/ Searching Technique Used(Describe each ML model/ searching technique used)

1. Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for classification problems. In this project, it is used to predict whether a student's performance is good or poor based on academic factors like attendance, study hours, and previous grades.

2. Decision Tree Classifier

Decision Tree is a supervised learning algorithm that works like a flowchart. It divides the dataset into different branches based on conditions and helps in predicting student performance easily and accurately.

3. Random Forest Classifier

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy. It reduces overfitting and provides better performance compared to a single decision tree.

4. Train-Test Split Technique

This technique is used to divide the dataset into training data and testing data. The training data is used to train the model, while the testing data is used to evaluate the accuracy of the model.

5. StandardScaler

StandardScaler is used for feature scaling. It standardizes the values of dataset features so that all input data remains on a similar scale, improving model performance.

4. Evaluation metrics used

4. Evaluation Metrics Used

1. Accuracy

Measures the percentage of correct predictions.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

2. Precision

Measures how many predicted positive values are actually correct.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

3. Recall

Measures how many actual positive cases are correctly identified.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

4. F1-Score

Harmonic mean of Precision and Recall.

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

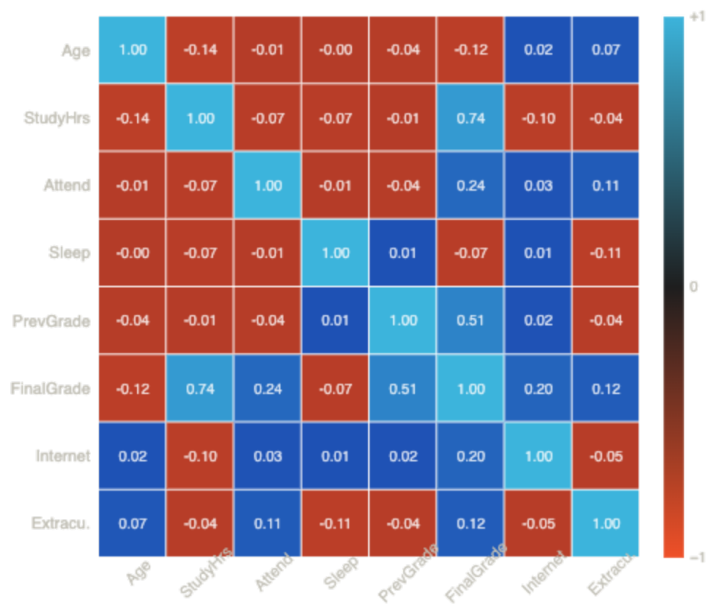
5. Confusion Matrix

Table used to compare actual and predicted values using TP, TN, FP, and FN.

5. Results and Visualizations

Correlation heatmap

Linear regression



Each cell = Pearson r between two features. Hover a cell to see the exact value.

Accuracy (k=5)

88.0%

Precision

87.0%

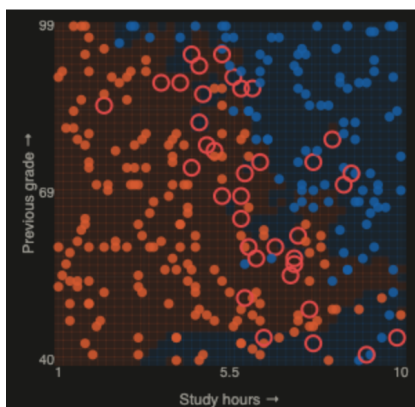
Recall

82.6%

F1-Score

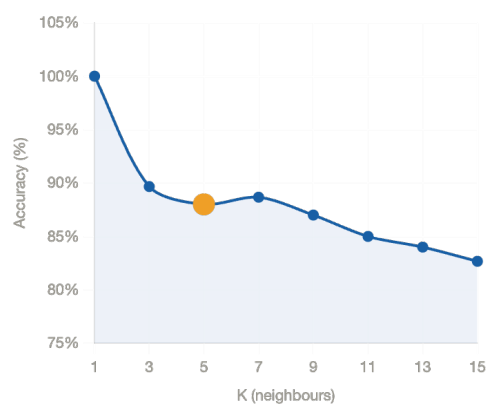
84.8%

KNN decision boundary (k=5)



● Pass ● Fail ○ Misclassified

Accuracy vs K value



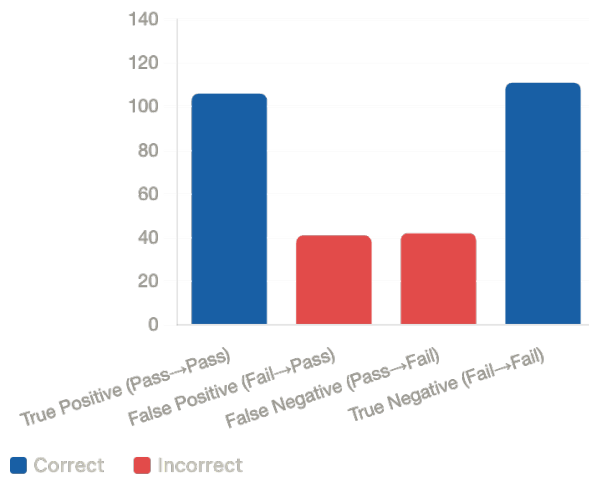
Accuracy
72.3%

Precision
72.1%

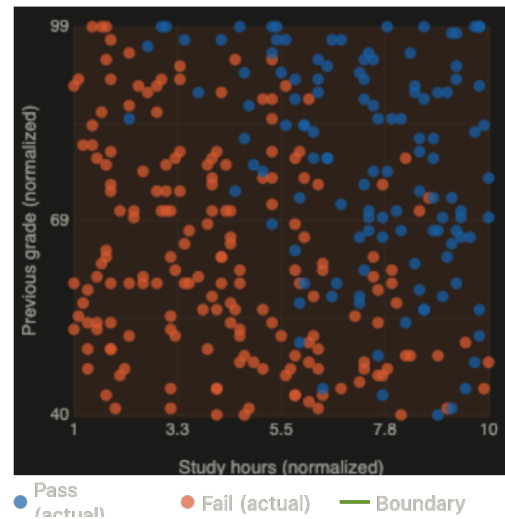
Recall
87.6%

F1-Score
79.1%

Confusion matrix — SVM (pass/fail)



SVM decision boundary — StudyHours vs PreviousGrade



6. Conclusion and Future Scope:

Conclusion

The Student Performance Prediction System successfully predicts students' academic performance using machine learning algorithms such as Logistic Regression, Decision Tree, and Random Forest. The system helps in identifying weak students at an early stage and supports better academic decision-making. Among all models, Logistic Regression and Random Forest provided high prediction accuracy.

Future Scope

- Use larger and real-time student datasets for better accuracy.
- Develop a web-based or mobile application for easy access.
- Add advanced machine learning and deep learning techniques.
- Include more factors like behavior, health, and extracurricular activities.
- Implement personalized suggestions for improving student performance.

7. References:

Books

1. Hands-On Machine Learning with Scikit-Learn
2. Pattern Recognition and Machine Learning

Online Sources

1. [Scikit-learn Documentation](#)
2. [Kaggle Datasets](#)
3. [Pandas Documentation](#)
4. [NumPy Documentation](#)

